LELME2150 - Thermal Cycles
Homework 4 - Steam Turbines

Introduction

The objective of this homework is to simulate a steam turbine cycle with regenerative feed-water heating and reheating, and to perform energy and exergy analyses. Through this work, students will be able to:

- Apply energy and exergy analyses to a multi-body steam turbine;
- Understand the roles of the drum, the closed feed-water heaters, and the reheater;
- Assess the influence of key design parameters on cycle efficiency;
- Connect simulation results to real industrial steam power plants.

This homework is mandatory! The **deadline** for this homework is **Sunday 15 November at 23:59**. It will be evaluated during an oral discussion for which you can enroll on the Moodle page. Before submitting, we advise you to test your code! A code that does not pass the basic test provided will receive a score of 0. Please submit your `Python` script and the 2-page report in the designated area on Moodle.

Tasks

1. Download the signature (`steam_turbine_group_xx.py`);
2. Rename the script with your group number (0x if the number is lower than 10);
3. Download the test code (`steam_turbine_test.py`);
4. Fill the signature with your own steam turbine model;
5. Prepare a **max 2-page report**. This report is a support document for the oral discussion: it should summarize your key results and findings rather than detail every step of your reasoning. We will go into depth together during evaluation. This report must at least include:
    - a summary of the most relevant and meaningful results and parametric analyses, **including the figures supporting the analyses**;
    - a conclusion where you suggest cycle/system improvements, discuss the main constraints and connect your results to real applications (coal power plant, combined heat and power, future of steam power, ...).

The final grade will be highly correlated to the quality of the parametric analyses and discussions.

Statement

Consider the steam turbine cycle as shown in Figure 1.

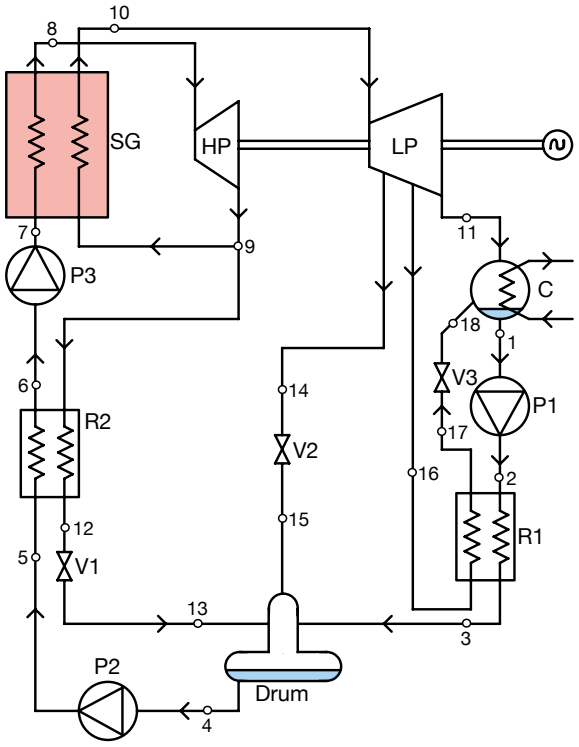


Figure 1: Steam turbine cycle investigated in this homework.

The cycle consists of **two turbine bodies** (HP and LP), a steam generator (SG) with **a reheating**, **three steam extractions**, and **three pumps** (P1, P2 and P3). The regeneration train comprises:

- one closed high-pressure feed-water heater, fed by steam extracted at the HP turbine exit (state 9);
- one open feed-water heater acting as a **drum** (bâche de dégazage), fed by steam extracted inside the LP turbine (state 15);
- one closed low-pressure feed-water heater, fed by steam extracted inside the LP turbine (state 16).

Modelling assumptions:

- The HP and LP turbines each have their own isentropic efficiency $\eta_{is,HP}$ and $\eta_{is,LP}$.
- A single mechanical efficiency η_{mec} applies to all turbine shafts.
- A single pump efficiency η_{pump} applies to all three pumps.
- A pressure-loss coefficient k_{heat} applies at the outlet of the steam generator and at the outlet of the reheater: $p_8 = k_{\text{heat}}\, p_7$ and $p_{10} = k_{\text{heat}}\, p_9$.

- Closed feed-water heaters are modelled with the pinch point method. We assume that states 12 and 17 are saturated liquid.
- The drum outlet is saturated liquid.
- The condenser outlet is saturated liquid.
- The minimum vapour quality at the LP turbine exit must satisfy $x_{11} \geq x_{\min}$.
- Combustion (or any external heat source) is modelled as a simple adiabatic heat transfer; no steam generator model is required.

The inputs and outputs of your model are fully described in the code signature. For background on the modelling methodology, please refer to the course textbook (Chapter 2, pp. 53–94).

Before submitting, please make sure that you follow these guidelines:

- Do not import any additional packages beyond those already provided.
- Ensure that when `display` is set to `True`, your code generates the T-s and h-s diagrams, and the energy and exergy pie charts of the cycle. When `display` is set to `False`, your code does not generate any plots.
- Ensure that the code produces no printed output.
- Ensure that all output variables are filled in. If the quality (x) of the state is not defined, set it to -1.
- Your submission must include exactly one PDF file and one code file.
- Do not modify the provided code template.